Notes: LISA Fourier-domain response for IMR waveforms

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We present an approach to process inspiral-merger-ringdown waveforms through the LISA response directly in the Fourier domain. We present a perturbative formalism based entirely on quantities defined from the Fourier-domain amplitude and phase of the waveform, investigate the magnitude of the higher-order corrections and show that including them gives a better control on the errors.

I. INTRODUCTION

In contrast with current ground-based detectors, which are typically sensitive to chirping binaries for less than a second, LISA signals may accumulating over months or years. The response of a LISA-type instrument is thus time-dependent [1]. The motion and change of orientation of the detector constellation along its orbit lead to significant time variability in the form of a modulation and a varying delay. These effects then convey information about the localization of the gravitational-wave source in the sky. The complicated detector response increase the computational cost of parameter-estimation analyses. To leverage the performance of state-of-the-art Fourier-domain IMR templates [2, 3], it would be useful to be able to process the signals through the time-dependent response of the dectector while staying in the Fourier domain.

The purpose of this paper is to introduce a formalism for efficiently processing signals through a time-domain modulation and delay within the Fourier domain, while retaining the compactness of a Fourier-domain amplitude and phase representation of the signals.

The problem of Fourier-domain modulations is also relevant in the context of waveform modelling for precessing systems, and our formalism can also be seen as an extension of the Shifted Uniform Asymptotics (SUA) introduced in [7, 8].

II. INTRODUCTION

A. Instrumental modulations and delays for LISA-type detectors

The response of a detector of the LISA type to an incident gravitational wave can be written in two different, equivalent forms, in terms of phase or frequency measurements. Here we will work with the second representation, which will prove more convenient for our purposes. Moreover, various notation and conventions have been used in the literature to label the spacecrafts and describe their orbits. We refer the reader to [11] for a comparative account on these various conventions. In this work, we will use the conventions of [11], which were also used in the Mock LISA Data Challenges (MLDC) [12].

We use a coordinate system centered on the solar system barycenter (SSB), and we denote the center of the constellation by O. We introduce the notation k for the propagation vector of the gravitational wave, which we denote by $h_{ij}^{\mathrm{TT}}(t)$ in transverse-traceless matrix form, as measured at the SSB (thus, at position p, $h_{ij}^{\mathrm{TT}}(t,p) = h_{ij}^{\mathrm{TT}}(t-k.p)$). We denote by p_A ($A=1,2,3$) the position of the spacecrafts and n_l the unit vectors of the three links, with the convention that n_3 points from 1 to 2.

Written in terms of the fractional laser frequency shifts between two spacecrafts, the elementary response of the detector reads [11, 13, 14]

$$\begin{aligned} y_{slr} &\equiv \frac{\nu_r - \nu_s}{\nu} \\ &= \frac{1}{2}\frac{n_l^i n_l^j}{1-k\cdot n_l}\left[h_{ij}^{\mathrm{TT}}(t-k\cdot p_s) - h_{ij}^{\mathrm{TT}}(t-k\cdot p_r)\right] . \end{aligned} \tag{1}$$

As detailed below, we will use a rigid instantaneous model, approximating the geometry of the constellation by a moving equilateral triangle (neglecting the flexing of the arms induced by corrections in the orbits), and evaluating all geometric factors in (1) at a single time t (neglecting pointing-ahead corrections).

The first- and second-generation TDI observables are then built as combinations of these basic building blocks, evaluated at delayed times. Since in our rigid approximation these delays will take a simple form in Fourier domain, we will focus in this paper on the y_{slr} observables. We will give in Sec. IV A more details on the response of LISA-type detectors, on the TDI observables, and on the approximations that enter the derivation of the basic response (1) above.

The structure of (1) is such that one can build the full signal from individual contributions of the form

$$s(t) = F(t)h(t+d(t)) . \tag{2}$$

Here, $h(t)$ represents one of the individual modes building the full gravitational wave signal (see Eq. (A2)), $d(t)$

represents the time-varying delays of the form $-k\cdot p_A(t)$, and $F(t)$ incorporates all the relevant geometric prefactors.

The functions $F(t)$ and $d(t)$ vary on a timescale of one year. When neglecting small non-periodic orbital perturbations (such as those caused by the influence of the Earth and of the other planets on the orbits), both $F(t)$ and $d(t)$ are periodic. It will also be useful to separate the delays in two types of terms: the first, $d_O = -k\cdot p_O$, relates the waveform at the SSB to the waveform at the center of the constellation O, whereas the second, d_c, represents the various delays between the spacecrafts of the constellation. For an orbit radius $R = 1\,\mathrm{au}$ and an armlength $L = 2.5\times10^6\mathrm{km}$ [15], $d_O \sim R/c \sim 500\mathrm{s}$ and $d_c \sim L/c \sim 8\mathrm{s}$.

B. Stationary phase approximation

As a preliminary step, we recall here an approximation widely used for similar purposes in treating gravitational waves signals emitted by inspiralling binaries, the Stationary Phase Approximation [5, 6, 16] (thereafter SPA). One first writes the time-domain signal in amplitude and phase form as $h(t) = a(t)e^{-2i\varphi(t)}$, where for gravitational wave signals φ will correspond to the orbital phase, with the orbital frequency being $\omega = \dot\varphi$. In order to keep close to the notation used in the gravitational wave literature, we introduced a factor of 2 in the phase of the wave, which is appropriate for the dominant 22 harmonic of the signal. The approximation then applies to signals verifying the conditions

$$\left|\frac{\dot a/a}{\omega}\right| \ll 1\,, \quad \left|\frac{\dot\omega}{\omega^2}\right| \ll 1\,, \quad \left|\frac{(\dot a/a)^2}{\dot\omega}\right| \ll 1\,. \tag{3}$$

Since the integral (A1) defining the Fourier transform is rapidly oscillatory unless the term $2\pi ft$ cancels the evolution of $-2\varphi(t)$, its support is well centered around the point of stationary phase. This determines the time-to-frequency correspondence in the stationary phase approximation, and leads to the definition of this time as an implicit function of frequency by the relation

$$\omega(t_f^{\rm SPA}) = \pi f\,. \tag{4}$$

For a chirping signal of increasing phase, $\omega > 0$ and $\dot\omega > 0$, there is a unique point of stationary phase, located in the positive frequency range $f > 0$. Using the conditions above, one can formally expand the signal around $t_f^{\rm SPA}$ to quadratic order in time, according to

$$\tilde h_{\rm SPA}(f) \simeq a(t_f^{\rm SPA})\exp\left[2i\pi ft_f^{\rm SPA} - 2i\varphi(t_f^{\rm SPA})\right] \cdot\int \mathrm{d}t\, e^{-i\dot\omega(t_f^{\rm SPA})(t-t_f^{\rm SPA})^2}\,. \tag{5}$$

Note that to be able to treat the amplitude as a constant in the integral above, we used the third condition in (3). The resulting complex Gaussian integral yields[1]

$$\tilde h_{\rm SPA}(f) = A_{\rm SPA}(f)e^{-i\Psi_{\rm SPA}(f)}\,, \tag{6a}$$

$$A_{\rm SPA}(f) = a(t_f^{\rm SPA})\sqrt{\frac{\pi}{\dot\omega(t_f^{\rm SPA})}}\,, \tag{6b}$$

$$\Psi_{\rm SPA}(f) = 2\varphi(t_f^{\rm SPA}) - 2\pi ft_f^{\rm SPA} + \frac{\pi}{4}\,. \tag{6c}$$

Ref. [16] evaluated the first correction to this approximation, within the context of post-Newtonian signals, and found that it can be considered as a term of the fifth post-Newtonian order, beyond the accuracy level of our current best models [17].

To understand the separation of timescales in the problem, it will be useful to have at hand the leading-order scaling laws for an inspiral (at the Newtonian order in the PN language). Although inaccurate for the purpose of waveform modelling, these Newtonian estimates will give useful orders of magnitude of the relevant timescales in the inspiral. For a binary with masses m_1, m_2, we define the total mass $m = m_1+m_2$ and the symmetric mass ratio $\nu = m_1m_2/m^2$. Introducing a time of coalescence t_c, the relations between the orbital frequency and phase and the time to coalescence $t_c - t$ are then given by

$$\omega(t) = \left[\frac{256\nu}{5c^5}(Gm)^{5/3}(t_c-t)\right]^{-3/8}\,, \tag{7a}$$

$$\varphi(t) = -\left[\frac{c^3}{Gm\nu^{3/5}}(t_c-t)\right]^{5/8}\,. \tag{7b}$$

As for the leading-order time-domain amplitude of the 22 mode, we have

$$a_{22}^{\rm N}(t) = \frac{2Gm\nu x}{Dc^2}\sqrt{\frac{16\pi}{5}}\,, \tag{8}$$

where we set $x = (GM\omega/c^3)^{2/3}$ and where D is the luminosity distance to the observer. For this Newtonian inspiral, applying the SPA gives for the 22 mode $\tilde h(f) = A_{\rm N}(f)e^{-i\Psi_{\rm N}(f)}$ with

$$A_N(f) = \frac{G^2m^2\pi}{Dc^5}\sqrt{\frac{2\nu}{3}}v^{-7/2}\,, \tag{9a}$$

$$\Psi_{\rm SPA}^{\rm N}(f) = \phi_0 - 2\pi ft_0 - \frac{3}{128\nu v^5}\,, \tag{9b}$$

where we used the notation $v = (Gm\pi f/c^3)^{1/3}$ ($x = v^2$ with the SPA correspondence (4)), and where t_0, ϕ_0 are constants. It is also customary to rewrite the above relations in terms of the chirp mass $\mathcal{M} \equiv m\nu^{3/5}$, which is the only mass combination arising in the phase at the leading PN order.

[1] Note that the expressions given here are valid for the 22 mode of the waveform.

In the Newtonian, low-frequency limit each of the combinations (3) assumed to be small in the SPA limit indeed vanishes at $\mathcal{O}((v/c)^5)$, but as the system approaches merger, this condition is no-longer satisfied. Near-merger however the treatment is not applicable, so that this is not a viable approach for binary systems beyond insprial. Nonetheless the SPA, has been a useful workhorse in many gravitational analysis involving frequency domain transfer functions. The usual approach is simply to replace any time dependencies appearing the transfer function using (4). We will later return to further consider the SPA treatment in the context of a more general formalism.

III. FOURIER-DOMAIN APPROACH TO MODULATIONS AND DELAYS

In this section we develop a Fourier-domain formalism for treating time-delays and temporally multiplicative signal transformations, where the period of primary signal oscillations is short compared to the timescale on which the modulating functions vary.

A. Fourier transform of a modulated and delayed signal

We begin by expression our delay and modulation function definitions using the Fourier transform (notice our unusual sign convention in (A1)),

$$\begin{aligned} h_d(t) &= h(t + d(t)) \\ &= \int \mathrm{d}f\, e^{-2i\pi f(t+d(t))}\tilde{h}(f)\,, \end{aligned} \tag{10}$$

and

$$\begin{aligned} \tilde{s}(f) &= F(t)h_d(t) \\ &= \int \mathrm{d}t\, e^{2i\pi ft} F(t) \int \mathrm{d}f'\, e^{-2i\pi f'(t+d(t))}\tilde{h}(f') \\ &= \int \mathrm{d}f'\, \tilde{h}(f-f') \int \mathrm{d}t\, e^{2i\pi f't} e^{-2i\pi(f-f')d(t)} F(t)\,. \end{aligned} \tag{11}$$

The last equation can then be rewritten as a kind of convolution with a frequency-dependent Kernel, according to

$$\tilde{s}(f) = \int \mathrm{d}f'\, \tilde{h}(f-f')\tilde{G}(f-f', f')\,, \tag{12}$$

where we introduced the frequency-dependent function of time $G(f,t)$, and its Fourier transform in the auxiliary frequency f', denoted by $\tilde{G}(f,f')$, as

$$G(f,t) = e^{-2i\pi f d(t)} F(t)\,, \tag{13a}$$

$$\tilde{G}(f,f') = \int \mathrm{d}t\, e^{2i\pi f't} e^{-2i\pi f d(t)} F(t)\,. \tag{13b}$$

In the absence of delays $d(t)$, as in the case of precessing binaries, the function G loses its frequency-dependence and becomes a modulation in the form of a function of time $F(t)$, and the result (12) above reduces to the usual convolution theorem for the Fourier transform.

A direct computation of (12) using (13b) will generally be computationally demanding, but when $\tilde{G}(f,f')$ has only limited support over f', we can seek an accurate but more efficient approximation. In our approach, two assumptions will be crucial for our approximations to be valid . The first is that the time scales entering the signal $h(t)$ are well separated from the ones entering the modulation and the delay, so that $\tilde{G}(f-f',f')$ has support limited to $|f'| < f'_{\mathrm{max}}$, with f'_{max} small compared the characteristic scale for the variation of the Fourier-domain signal. If additionnally the original signal has a smooth amplitude and phase in the Fourier-domain, slowly varying over the scale f'_{max}, a local treatment in f' of the convolution in (12) will be possible.

Notice that these approximations are not exactly the same as the ones usually made to justify the stationary phase approximation for the treatment of the Fourier transform of the inspiral part of the signal. Indeed, we will exploit the smoothness of the Fourier-domain $\tilde{h}(f)$ to be able to cover the merger-ringdown part of the signal h, where the SPA itself breaks down.

B. Leading order: the local-in-frequency approximation

Under the assumptions above, it is possible to see the convolution in the f' variable in (12) as being a local operation, allowing us to expand the signal $\tilde{h}(f-f')$ around f. A first leading-order approximation, which could be called the local approximation for reasons that will appear clear below, is to discard the f' dependence in the first argument of $\tilde{g}(f-f',f')$, to keep only the leading order in the Fourier-domain amplitude, and to expand the Fourier-domain phase to the first order, according to

$$\tilde{h}(f-f') \simeq A(f)\exp\left[-i\left(\Psi(f) - f'\frac{\mathrm{d}\Psi}{\mathrm{d}f}\right)\right]\,, \tag{14}$$

Here and in the rest of the paper, derivatives with respect to the frequency will always be evaluated at f. Notice that, because of the freedom of adding a linear term to $\Psi(f)$ by simply shifting the signal in time, no assumption can be made on the smallness of the first derivative of the phase, and this is really a leading order approximation.

Plugging this relation in (12), we obtain

$$\begin{aligned} \tilde{s}(f) &= \tilde{h}(f) \int \mathrm{d}f'\, \exp\left[if'\frac{\mathrm{d}\Psi}{\mathrm{d}f}\right] \tilde{G}(f,f') \\ &= \tilde{h}(f) G\left(f, -\frac{1}{2\pi}\frac{\mathrm{d}\Psi}{\mathrm{d}f}\right)\,, \end{aligned} \tag{15}$$

which motivates the introduction of an effective time,

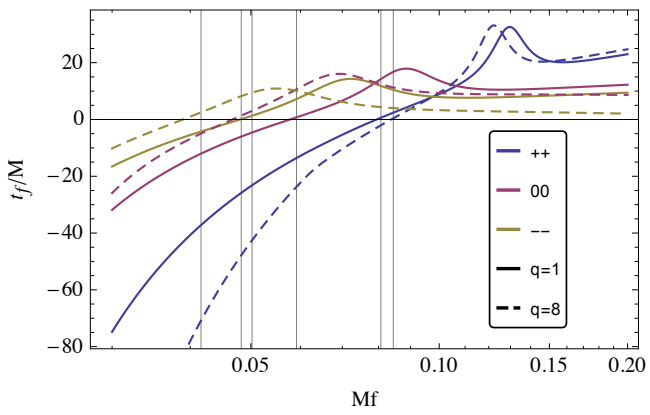


FIG. 1. Time-to-frequency correspondence t_f, as defined directly from the phase of the Fourier-domain signal as in (16), in geometric units. We show only the high-frequency part of the signal, corresponding to the merger region. The PhenomD waveforms have been aligned such that the time-domain amplitude (obtained by an IFFT) peaks at $t = 0$. The two aligned spins are equal, with components of 0.95 (++, blue), 0 (00, red), and -0.95 ($--$, yellow), for mass ratios $q = 1$ (full line) and $q = 8$ (dashed). The vertical lines shows the time-domain instantaneous frequency at the peak $\omega^{22}_{\rm peak}/(2\pi)$, again measured from an IFFT, and the fact that the curves t_f do not pass exactly by the crossing with the $t = 0$ horizontal line reflects the fact that the link between time-domain frequency and Fourier-domain f is not accurate anymore in the merger part of the signal. Note that t_f does not go beyond $\sim 30M$ after the peak, and is not monotonous.

seen as a function of frequency,

$$t_f \equiv -\frac{1}{2\pi}\frac{\mathrm{d}\Psi}{\mathrm{d}f}\,, \tag{16}$$

This definition is a generalization of the time-to-frequency correspondence at the heart of the SPA, but it is worth noting that here it is defined entirely from the Fourier-domain waveform, and independently of the SPA being valid or not for the underlying signal $\tilde{h}$. As such, this time-of-frequency function also extends to the merger-ringdown part of the signals. Fig. 1 shows the behaviour of the time function t_f around merger for six example waveforms, for mass ratios $q = 1$ and $q = 8$ and for aligned spin components $\chi = 0.95, 0., -0.95$. In particular, one should note that t_f is not monotonously increasing with frequency anymore when reaching in the high-frequency part of the waveform, corresponding to the ringdown. This would forbid an unambiguous definition of an inverse frequency-of-time correspondence $f(t)$.

Using this notation, we have simply

$$\mathcal{T}_{\rm local}(f) = G(f, t_f)\,. \tag{17}$$

The interpretation of this approximation is straightforward: the signal is simply multiplied by the response function evaluated at the time t_f. For precessing binaries, this corresponds (but is not quite identical to) the approximation used for the PhenomP approximant [9]. Additionally, when delays are present, the frequency-dependence of the response is evaluated directly at f.

It is worth noting that a shift in time of the time-domain signal will, by construction, be appropriately propagated to (17). If $h'(t) = h(t - \Delta t)$ [but the modulation and delays are left unchanged (which is relevant for space-based detectors, not for precessing binaries)], $\Psi'(f) = \Psi(f) - 2\pi f \Delta t$, so that

$$\tilde{s}'_{\rm local}(f) = \tilde{h}'(f) G(f, t_f + \Delta t)\,. \tag{18}$$

This observation also justifies that the first derivative of the Fourier-domain phase cannot, in general, be considered a small quantity, as it can take arbitrary values for arbitrary shifts in time of the signal. The leading order approximation corresponds to neglecting the curvature in $\Psi(f)$, and yields a result (17) where the passage to the Fourier domain is done by simply using a time-to-frequency correspondence (16), which motivates the name "local approximation".

C. Taylor expansion in the Fourier domain

We now extend our approach beyond the local approximation. We will give a formal derivation of the general formula for the higher-order corrections in (12), and then specialize it to the special cases of interest.

We will formally Taylor-expand $\tilde{h}(f - f')$ in f' as follows:

$$\Psi(f - f') = \Psi(f) + 2\pi f' t_f + \sum_{p\geq 2}^{N} \frac{(-1)^p}{p!} f'^p \frac{\mathrm{d}^p \Psi}{\mathrm{d}f^p}\,, \tag{19a}$$

$$A(f - f') = A(f) \sum_{q\geq 0} \frac{(-1)^q}{q!} f'^q \frac{1}{A}\frac{\mathrm{d}^q A}{\mathrm{d}f^q}\,, \tag{19b}$$

$$\tilde{G}(f - f', f') = \sum_{r\geq 0} \frac{(-1)^r}{r!} f'^r \frac{\partial^r}{\partial f^r} \tilde{G}(f, f')\,. \tag{19c}$$

This expansion will only be usable in practice when we will be able to keep only a few terms in each of these expansions. We therefore limited the expansion in Ψ to a finite order N, to keep the formula readable. Notice that we expand $\tilde{G}(f - f', f')$ in f' only in its first argument. We can also interchange the f-derivatives of g with its Fourier transform.

If we further expand the exponentials to obtain a pure f'-expansion, by applying the formal derivative rule

$$\int \mathrm{d}f'\, f'^n \frac{\partial^m}{\partial f^m}\tilde{G}(f,f')e^{-2i\pi f' t_f} = \frac{1}{(-2i\pi)^n}\left(\frac{\partial^m}{\partial f^m}\frac{\partial^n}{\partial t^n}G\right)(f,t_f)\,, \tag{20}$$

we arrive at the formal combined expansion

$$\mathcal{T}(f) = \sum_{q\geq 0}\sum_{r\geq 0}\sum_{k_2\geq 0}\cdots\sum_{k_N\geq 0}\frac{(-1)^q}{q!}\frac{(-1)^r}{r!}\prod_{p=2}^{N}\frac{1}{k_p!}\left(\frac{(-1)^p(-i)}{p!}\frac{\mathrm{d}^p\Psi}{\mathrm{d}f^p}\right)^{k_p}\frac{1}{A}\frac{\mathrm{d}^q A}{\mathrm{d}f^q}\frac{1}{(-2i\pi)^m}\left(\frac{\partial^r}{\partial f^r}\frac{\partial^m}{\partial t^m}G\right)(f,t_f)\,, \tag{21}$$

where $m = p + r + 2k_2 + \cdots + Nk_N$.

This cumbersome expansion will not be used as such, but takes a more intuitive form for special cases. Of particular interest for us will be the case where we keep only the first term of the series in (19a), and expand the resulting exponential, which yields

$$\mathcal{T}_{\mathrm{phase}}(f) = \sum_{p\geq 0}\frac{1}{p!}\left(\frac{i}{8\pi^2}\frac{\mathrm{d}^2\Psi}{\mathrm{d}f^2}\right)^p\left(\frac{\partial^{2p}}{\partial t^{2p}}G\right)(f,t_f)\,. \tag{22}$$

Indeed, this correction will be shown in Sec. III E to be closely related to the previous approach of [8]. The expansion shows that the transfer function is signal-dependent through the second derivative of the phase Ψ. Similarly, the expansion (19b) gives

$$\mathcal{T}_{\mathrm{amp}}(f) = \sum_{q\geq 0}\frac{1}{(2i\pi)^q q!}\frac{1}{A}\frac{\mathrm{d}^q A}{\mathrm{d}f^q}\left(\frac{\partial^q}{\partial t^q}G\right)(f,t_f)\,. \tag{23}$$

If we expand only the frequency-dependence of G as in (19c), we obtain

$$\mathcal{T}_{\mathrm{delay}}(f) = \sum_{r\geq 0}\frac{1}{(2i\pi)^r r!}\left(\frac{\partial^r}{\partial f^r}\frac{\partial^r}{\partial t^r}G\right)(f,t_f)\,, \tag{24}$$

which interestingly looks like a Taylor expansion of G, but with joint derivatives in frequency and time. This time, the expansion is signal-independent as it only depends on the function G and not on A, Ψ.

In the following, we will rewrite in a more convenient form the formal Taylor expansions for the quadratic phase effects and the delay effects. As will be explained in Sec. III H, we will use these Taylor expansions to build error measures designed to keep track of which of these three types of corrections is important.

D. Signal-dependent timescales

Considering (22) it is natural to define a new timescale, as a function of frequency, as

$$T_f^2 = \frac{1}{4\pi^2}\left|\frac{\mathrm{d}^2\Psi}{\mathrm{d}f^2}\right|\,. \tag{25}$$

The interpretation of this timescale for the inspiralling part of the signal, when the SPA applies, will be discussed below. Whe can use this notation to rewrite (22) as

$$\mathcal{T}_{\mathrm{phase}}(f) = \sum_{p\geq 0}\frac{(-i\epsilon)^p}{2^p p!}T_f^{2p}\left(\frac{\partial^{2p}}{\partial t^{2p}}G\right)(f,t_f)\,, \tag{26}$$

where $\epsilon = -\mathrm{sgn}(\mathrm{d}^2\Psi/\mathrm{d}f^2)$ is 1 in the inspiral.

The expansion in amplitude corrections (23) also leads to the natural introduction of a set of other timescales related to the successive derivatives of the amplitude. In an analogous manner to the definition (25) of the timescale T_f, we can define

$$(T_{An})^n \equiv \frac{1}{(2\pi)^n}\frac{1}{A(f)}\left|\frac{\mathrm{d}^n A}{\mathrm{d}f^n}\right|\,, \tag{27}$$

where we included an absolute value to accomodate the possible sign changes in the right-hand side. Although the definition above is for a generic $n \geq 0$, in practice only the first few of these timescales will be relevant. In this paper we will use only the first two, T_{A1} and T_{A2}.

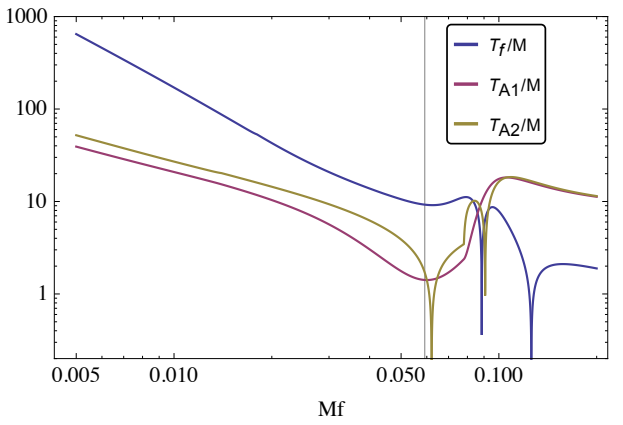


FIG. 2. Fourier-domain amplitude and phase timescales, as defined in (25) and (27), in geometric units for an equal-mass, non-spinning system. The hierarchy of these timescales is roughly the same for higher mass ratio and higher spin systems. The time-domain frequency at merger is represented by the vertical line. The derivatives from which these timescales are built encounter zero-crossings and change sign in the high-frequency range.

Of particular interest is the relation between out formal results and the SPA. For non-precessing gravitational wave signals, the SPA is a very good approximation through the inspiralling phase [16]. However, it is important to note that the approximation breaks down when reaching the merger-ringdown part of the signal, as the rapidly varying features of the merger break the locality of the time-to-frequency relation (4).

Taking the first derivative of the defining relation (4) gives

$$t_f = -\frac{1}{2\pi}\frac{\mathrm{d}\Psi_{\rm SPA}}{\mathrm{d}f}\,, \tag{28}$$

which means that our generalized definition (16) for the effective time t_f coincides with $t_f^{\rm SPA}$ in the inspiral phase of the signal, where the SPA is applicable. Similarly, if we define

$$T_f^{\rm SPA} \equiv \frac{1}{\sqrt{2\dot{\omega}(t_f)}}\,, \tag{29}$$

we have

$$\left(T_f^{\rm SPA}\right)^2 = -\frac{1}{4\pi^2}\frac{\mathrm{d}^2\Psi_{\rm SPA}}{\mathrm{d}f^2}\,, \tag{30}$$

where $\mathrm{d}^2\Psi/\mathrm{d}f^2 < 0$ in the SPA with our sign conventions. Comparing with (25) above, we see that our generalized time scale T_f also reduces to $T_f^{\rm SPA}$ when the SPA applies. However, its definition extends to the merger-ringdown part of the signal, and we have to include an absolute value to allow for a possible change of sign of $\mathrm{d}^2\Psi/\mathrm{d}f^2$ in that regime, as shown in Fig. 2.

It is useful, to estimate the magnitude of the corrections with respect to the leading order, to give an estimate for these timescales based on the Newtonian expressions (9). This gives for the Newtonian radiation-reaction and amplitude timescales:

$$T_f^{\rm N} = \frac{1}{8}\sqrt{\frac{5}{3\nu}}\frac{Gm}{c^3}v^{-11/2}\,, \tag{31a}$$

$$T_{A1}^{\rm N} = \frac{7}{6}\frac{1}{2\pi f}\,. \tag{31b}$$

One can check explicitly that this expression for T_f agrees with (29). With a simple power-law amplitude as in (9a), all higher-order amplitude timescales are also simply proportional to $1/f$ and differ from the first only by their numerical factor.

E. Quadratic term in the phase and relation to the SUA

In this Section, we focus on corrective terms coming from the second-derivative of the phase, given in (22). If we restrict to the case of a pure modulation, $d = 0$ and $G(f,t) = F(t)$ [go back to more general $G(f,t)$?], we can relate our result (26) to the approach of Ref. [8], where the authors extend the SPA in a formalism called the shifted uniform asymptotic expansion (SUA). The derivation includes an expansion of the modulation in Bessel functions, a study of the displacement of the stationary phase point induced by the precession, and a resummation of the result in the form of time derivatives. The main result of [8], their equation (34), reads exactly like (26) with the identifications $\tilde{H}_{corr}(f) \to \mathcal{T}(f)$ for the transfer function, $T \to T_f$ for the radiation-reaction timescale and $e^{-i\delta\phi} \to F$ for the modulation function (which is restricted in their framework to include only the phase of the modulation, its amplitude being treated jointly with the amplitude of the signal). Thus, we are able to make a direct correspondence between the treatment of [8] and our framework, which provides us with a simple and alternative derivation of their result. Namely, the result of [8] corresponds to the approximation (19a), where in the Fourier-domain convolution the phase of the signal is expanded to quadratic order and the amplitude is not expanded.

The authors of [8] proposed a resummation scheme. Indeed, the result (26) looks like a symmetrized Taylor expansion, except for the factors i^p and $1/p!$ instead of $1/(2p)!$. After having truncated the sum at some finite N, one can write (following [8])

$$\begin{aligned}\mathcal{T}(f) &\simeq \sum_{p=0}^{N}\frac{(-i\epsilon T_f^2)^p}{2^p p!}\partial_t^{2p}F(t_f)\\ &\simeq \sum_{p=0}^{N}\sum_{k=0}^{N} a_{N,k}\frac{(kT_f)^{2p}}{(2p)!}\partial_t^{2p}F(t_f)\\ &\simeq \frac{1}{2}\sum_{k=0}^{N} a_{N,k}\left(F(t_f+kT_f)+F(t_f-kT_f)\right)\,,\end{aligned} \tag{32}$$

where one applies a truncation approximation at the first and last stage. The second equality holds provided that the complex coefficients $a_{N,k}$ are a solution of the $N+1$-dimensional linear system [8]

$$(-i\epsilon)^p(2p-1)!! = \sum_{k=0}^{N} a_{N,k}k^{2p} \quad \text{for } p = 0,\ldots,N\,. \tag{33}$$

The two solutions for the stencil coefficients in the two cases $\epsilon = \pm 1$ are simply related by a complex conjugation.

An immediate advantage of this reformulation is its improved numerical stability. For the signal of precessing binaries, although one should expect an ideal scaling $\partial_t^p F \sim \Omega^p F$ with Ω the precession frequency, it can be extremely hard to control numerical derivatives of F (which in any realistic model will be only provided as sampled at discrete times) well enough to enforce this scaling. Here, however, one simply evaluates the original smooth function F at shifted times.

Note that the choice of the samples $t_f \pm kT_f$ to approximate the successive derivatives of the expansion by finite

differences, even if simple and natural, is by no means unique. In principle, one could have made any ansatz of samples for the series expansion to be matched on. One cannot expect all choices to work equally well when applied to noisy data, and the problems of the numerical size of the coefficients and of the overall numerical stability of the scheme must be investigated numerically. We leave a more complete investigation of the asymptotics and numerical stability of this resummation approach for future work.

F. Relation to the Fresnel transform

We can give an alternative interpretation of (26) and of Sec. III E above by using an integral approach. For simplicity of notation, here we will again keep to the case of precessing binaries, with only a modulation F and no delays, $d = 0$. We will reintroduce the delays below in Sec. III G. To obtain (26), we expanded fully the exponential as a Taylor series. If it is not expanded, however, we have

$$\tilde{s}(f) \simeq \tilde{h}(f) \int \mathrm{d}t\, F(t) \int \mathrm{d}f'\, e^{2i\pi f'(t-t_f)} \exp\left[2i\pi^2 \epsilon f'^2 T_f^2\right] , \tag{34}$$

where we recall that $\epsilon = -\mathrm{sgn}(\mathrm{d}^2\Psi/\mathrm{d}f^2)$. The integral on f' is a simple complex Gaussian integral, and results in a complex Gaussian integral over time where we recognize a Fresnel transform of the function F. We introduce for the Fresnel transform the notation

$$\mathcal{F}_\tau[F](t_0) \equiv \frac{e^{i\frac{\pi}{4}}}{\sqrt{2\pi\tau}} \int \mathrm{d}t\, \exp\left[-\frac{i}{2}\left(\frac{t-t_0}{\tau}\right)^2\right] F(t) , \tag{35}$$

together with the additional notation

$$\mathcal{F}^\epsilon_\tau[F](t_0) \equiv \begin{cases} \mathcal{F}_\tau[F](t_0) & \text{if } \epsilon = 1 \\ \mathcal{F}_\tau[F^*](t_0)^* & \text{if } \epsilon = -1 \end{cases} \tag{36}$$

to accomodate for the possible sign change represented by ϵ. The integral (34) then reads

$$\tilde{s}(f) = \tilde{h}(f) \mathcal{F}^\epsilon_{T_f}[F](t_f) . \tag{37}$$

The Fresnel transform is an integral transform that shares some properties with the Fourier transform. It intervenes in particular in the theory of diffraction. Contrarily to the Fourier transform, however, it is localized in the sense that the part of the integral that is centered around t_0 contributes predominantly, due to the cancelling oscillations far from t_0. Note also that only the part of the function F that is symmetric about t_0 contributes to the integral in (35).

The parameter τ of the transform plays the role of a focusing parameter, determining how local the transform is. In the limit $\tau \to 0$, fast oscillations away from the central value t_0 will cancel out, leading to the integral taking the value $F(t_0)$. For large values of τ, however, the integral (35) has an extended support. Transposed in terms of the timescale T_f, since we have seen that T_f can be interpreted as the radiation reaction timescale in the SPA regime, this means that a faster-chirping signal (T_f small) will have a Fresnel transform that is more focused, whereas a slower-chirping signal (T_f large) will have a Fresnel transform that is more extended. How much the Fresnel transform is focused in time is to be put in relation with how fast the function $F(t)$ in the integrand is varying. In the case of signals from precessing binaries, the precession timescale evolves as the system gets closer to merger. In the case of the response of a LISA-like detectors, the modulation evolves with a timescale of one year, as will be detailed in Sec.IV B.

Although it is interesting to be able to relate our results to a known transformation used in signal processing, whether this opens avenues for more efficient computation methods remains to be investigated, for instance by performing a decomposition of the modulation F in chirplets, which have analytically computable Fresnel transforms. Note that in the result (37), both the scale T_f and the central time t_f are functions of the frequency f.

As a verification, upon performing a formal Taylor expansion in time of $F(t)$ around t_f in the above integral and integrating term by term the result[2], one readily recovers our previous result (26). Now, the resummation of (32) can be rephrased as a quadrature rule for computing the Fresnel integral (35), according to

$$\mathcal{F}^\epsilon_\tau[F](t_0) \simeq \frac{1}{2} \sum_{k=0}^{N} a^\epsilon_{N,k} \left(F(t_0 + k\tau) + F(t_0 - k\tau)\right) , \tag{38}$$

with $a^1_{N,k} = a_{N,k}$ and $a^{-1}_{N,k} = a^*_{N,k}$. Formally, if one allows for polynomial integrands in (35) (which can only be given sense by introducing a regularization), Eq. (38) amounts to the building a quadrature rule for the particular choice of nodes $t_0 \pm k\tau$, which is exact (with a regularization) if F is a symmetric polynomial of degree $\leq 2N$. Note however that this quadrature rule does not fall into the class of Gaussian quadratures, which provide N-point rules that give an exact representation of the integral for polynomials of degree $2N-1$. Due to the complex character of the kernel in (35), the induced inner product is not positive definite and some key properties are lost, such as the construction of the optimal Gaussian nodes from the zeros of the orthonormalized polynomials.

[2] Note that the intermediate integrals produced by such a Taylor expansion are formally divergent. One can regularize them for instance by introducing a small imaginary part in τ that is sent it to 0 at the end of the computation.

G. Treatment of the delays

In the case of a LISA-type detector response, the presence of the delay $d(t)$ is responsible for the frequency-dependence of the kernel G introduced in (13). We will show below that the higher-order correction in the delay can become non-negligible at high frequencies, as measured by the magnitude of the first correction in (24). This is especially true for the orbital delay related to the motion of the whole detector constellation around the Sun, due to the large baseline for this delay.

Here we give an alternative treatment of the delays which is an alternative to the mere Taylor expansion of (24). It is useful to look first at the case where one neglects the signal-related amplitude and phase corrections as given in (23) and (26), but keeps the frequency-dependence in the convolution kernel in (13). The response then depends on the signal only by the time-to-frequency correspondence and reads

$$\tilde{s}(f) = \tilde{h}(f) \int \mathrm{d}t\, F(t) e^{-2i\pi f d(t)} \int \mathrm{d}f'\, e^{2i\pi f'(t+d(t)-t_f)}\,, \tag{39}$$

where we recall that we defined $t_f \equiv -1/(2\pi)\mathrm{d}\Psi/\mathrm{d}f$. This motivates the introduction of the delayed time function $t_d : t \mapsto t + d(t)$, and of the reciprocal function t_d^{-1}, and of a modified time-to-frequency correspondence $t_f^d = t_d^{-1}(t_f)$, defined implicitly by

$$(t + d(t) - t_f)|_{t=t_f^d} = 0\,. \tag{40}$$

The above integral can then be formally computed as

$$\begin{aligned}\tilde{s}(f) &= \tilde{h}(f) \int \mathrm{d}t\, F(t) e^{-2i\pi f d(t)} \delta(t + d(t) - t_f) \\ &= \tilde{h}(f) F(t_f^d) \frac{e^{-2i\pi f d(t_f^d)}}{1 + \dot{d}(t_f^d)}\end{aligned} \tag{41}$$

For the standard LISA configuration, we have for the orbital delays the scaling $d_O \sim R \sim 500s$, for the constellation delays the scaling $d_c \sim L \sim 8s$ (with $c = 1$ and ignoring the dependence on angular factors). Since the motion of the constellation is periodic with a period of one year and $\Omega_0 = 2 \times 10^{-7}\mathrm{rad}.s^{-1}$, we have $\dot{d}_O \sim \Omega_0 R/c \sim 10^{-4}$ and $\dot{d}_c \sim \Omega_0 L/c \sim 1.7 \times 10^{-6}$. The smallness of the dimensionless quantity $\dot{d} \ll 1$ (and of its subsequent derivatives) will allow us to treat it perturbatively with a very good approximation, and shows also that the function t_d is univalued and that there is no ambiguity in defining its reciprocal t_d^{-1}.

By treating $\dot{d}$ as a perturbation and keeping only first-order terms, we obtain for the delayed time reciprocal function

$$\begin{aligned} t_d^{-1}(t) &\simeq t - d(t)(1 - \dot{d}(t))\,, \\ d(t_f^d) &\simeq d(t_f)(1 - \dot{d}(t_f))\,. \end{aligned} \tag{42}$$

Now, the most relevant correction in (41) comes from the phase factor at high frequencies, where the factors $2\pi f d_O$ and $2\pi f d_c$ give a magnification by a factor of, respectively, 3.10^3 and 10^2 at 1 Hz. Ignoring the other corrections, we thus arrive at the approximate form for the transfer function

$$\mathcal{T}(f) \simeq F(t_f) \exp\left[-2i\pi f d(t_f)(1 - \dot{d}(t_f))\right]\,. \tag{43}$$

Notice that this leading-order correction in the treatment of the delays affects purely the phase of the signal.

Next, we consider the case where the quadratic phase correction is kept as well, as in Sec. III F. Keeping as before only the first-order terms in $\dot{d}$ (and neglecting its higher derivatives), we can write

$$\begin{aligned}\tilde{s}(f) &\simeq \tilde{h}(f) \int \mathrm{d}t\, F(t) e^{-2i\pi f d(t)} \int \mathrm{d}f'\, \exp\left[2i\pi\epsilon T_f^2 f'^2 + 2i\pi f'(t + d(t) - t_f)\right] \\ &\simeq \tilde{h}(f) \frac{e^{i\epsilon\frac{\pi}{4}}}{\sqrt{2\pi}T_f} \int \mathrm{d}\tau\, \frac{F(\tau - d(\tau))}{1 + \dot{d}(\tau)} e^{-2i\pi f d(\tau)(1-\dot{d}(\tau))} \exp\left[-\frac{i\epsilon}{2}\frac{(\tau - t_f)^2}{T_f^2}\right]\,,\end{aligned} \tag{44}$$

where we used a change of variable $\tau = t_d(t)$. We see that the result can again be expressed as a Fresnel transform. When considering amplitude corrections as well, as in (23), additional powers of f' can be translated as time derivatives with respect to the variable τ after performing the change of variables. This produces our most general result given in Eq. (45) below.

H. Executive summary of the formalism

In this Section, we gather our previous results for the convenience of the reader. For a modulation F and delay d, so that $s(t) = F(t)h(t+d(t))$, we defined the associated function $G(f,t) = F(t)e^{-2i\pi f d(t)}$ and we obtained the transfer function $\mathcal{T}(f) = \tilde{s}(f)/\tilde{h}(f)$:

$$\mathcal{T}(f) = \sum_{k\geq 0} \frac{(-i)^k}{k!} (T_{Ak})^k \mathcal{F}^{\epsilon}_{T_f} \left[\frac{\mathrm{d}^k}{\mathrm{d}\tau^k} \left(\frac{F(\tau - d(\tau))}{1+\dot{d}(\tau)} e^{-2i\pi f d(\tau)(1-\dot{d}(\tau))} \right) \right] (t_f) \,. \tag{45}$$

In each term of this series, the Fresnel transform $\mathcal{F}^{\epsilon}_{T_f}$ can in turn be approximated by using the stencil formula (38). The timescales T_f and T_{Ak} were defined in (25) and (27), and the generalized time-to-frequency function t_f was given in (16). The delays d are present only in the LISA context, while in the context of precessing binaries we only have to consider the modulation function F. In practice, only the first few of the terms in this last remaining series expansion are relevant. We will investigate several orders of approximation, combining corrections from the phase, amplitude and delays, and explore which ones are relevant for a given level of accuracy. We will use symbols of the form $\{N, A, d\}$ to indicate the order of the stencil in (38), the maximal order of the amplitude correction included, and the inclusion or not of the delay correction.

In the LISA context, due to the smallness of the corrections we will only go up to $k = 1$ in (45), and in computing the remaining time derivatives in (45) we will neglect the second and higher derivatives. We will also use $F(\tau - d(\tau)) \simeq (F - d\dot{F})(\tau)$. This gives concretely:

$$\{N, A:0, d:0\}: \ \mathcal{T}(f) = \mathcal{F}^{\epsilon}_{T_f} \left[F e^{-2i\pi f d} \right] (t_f) \,, \tag{46a}$$

$$\{N, A:1, d:0\}: \ \mathcal{T}(f) = \mathcal{F}^{\epsilon}_{T_f} \left[\left(F - iT_{A1} \left(\dot{F} - 2i\pi f \dot{d} \right) \right) e^{-2i\pi f d} \right] (t_f) \,, \tag{46b}$$

$$\{N, A:0, d:1\}: \ \mathcal{T}(f) = \mathcal{F}^{\epsilon}_{T_f} \left[\frac{F - d\dot{F}}{1+\dot{d}} e^{-2i\pi f d(1-\dot{d})} \right] (t_f) \,, \tag{46c}$$

$$\{N, A:1, d:1\}: \ \mathcal{T}(f) = \mathcal{F}^{\epsilon}_{T_f} \left[\frac{1}{1+\dot{d}} \left(F - d\dot{F} - iT_{A1} \left(\dot{F} - 2i\pi f \dot{d} \right) \right) e^{-2i\pi f d(1-\dot{d})} \right] (t_f) \,. \tag{46d}$$

In the following, it will be convenient to introduce error estimates built from the Taylor-like series (22), (23) and (24). We simply define this error estimates at a certain level of approximation as the magnitude of the first term ignored in the original Taylor series. Thus we define

$$\epsilon_{\Psi 2} \equiv \frac{1}{2} T_f^2 \left| \frac{1}{G} \partial_{tt} G \right| \,, \tag{47a}$$

$$\epsilon_{A1} \equiv T_{A1} \left| \frac{1}{G} \partial_t G \right| \,, \tag{47b}$$

$$\epsilon_{A2} \equiv \frac{1}{2} T_{A2}^2 \left| \frac{1}{G} \partial_{tt} G \right| \,, \tag{47c}$$

$$\epsilon_{d} \equiv \frac{1}{2\pi} \left| \frac{1}{G} \partial_{tf} G \right| \,, \tag{47d}$$

where the function G and its derivatives are evaluated at (f, t_f). As we will detail below, in the presence of delays these derivatives do not obey the simple scaling $\partial_t^n \to \Omega_0^n$, with Ω_0 the characteristic frequency for the time variablility in the modulation and delay. This is due to the presence of additional dimensionless delay terms of the form $2\pi f d$, which can be larger than 1.

IV. APPLICATION TO THE RESPONSE OF LISA-TYPE DETECTORS

A. The response model

In this Section, we expand Sec. II A and detail more the model that we use for the response of a LISA-like detector, together with the assumptions used and their limitations.

As argued in [18], since we focus on the gravitational-wave contribution to the basic observable, we can use a simplified model for the response, ignoring corrections that are crucial from the point of view of noise cancellations. Our aim is indeed to assess the accuracy of a direct Fourier-domain treatment of the response, thus focusing on the separation of timescales in the problem, keeping in mind that corrections can be added later without affecting the conclusions of the present analysis.

The frequency-shift response for a single link (1) was derived in Ref. [13]. Several assumptions enter the result as written in (1): (i) effects of the order v/c are neglected, including for instance the special relativistic Doppler effect created by the relative speeds of the spacecrafts on their orbits (ii) the propagation is assumed to take place in a flat spacetime, perturbed only by the gravitational wave; thus the gravitational redshift as well as the deflection of light created by the gravitational potential of

the Sun is ignored (iii) all geometric factors are evaluated at a single time, whereas one should consider the beam as propagating from the position of the first spacecraft at the time of emission to the position of the second scapecraft at the time of reception, leading to a pointing-ahead effect.

Additionally, we limit ourselves to a rigid model for the detector, namely we assume that the constellation remains in an equilateral configuration with fixed armlengths $L = 2.5 \times 10^6$km, as proposed in [15]. This simplified orbits neglect (iv) the effect of excentricity of the individual Keplerian orbits, which introduces corrections at the level e^2 (v) the effect of gravitational perturbations coming from other celestial bodies, such as the Earth, the quadrupole of the Sun, and the other planets.

Note that although we can neglect all the effects (i)-(v) for our present study, keeping track of these corrections is crucial for the purpose of noise cancellations, and led to the development of new generations of TDI observables.

Since the orbital delay due to the orbit around the Sun is common to all y_{slr} observables, we will decompose the full instrument response [11, 13, 14] in two steps. The first step consists simply in applying the varying time delay to bring the signal from the SSB to the center of the LISA triangular constellation. For $h^{\rm TT}$ the transverse-traceless gravitational waveform in matrix form, we write this orbital time delay as

$$h_O(t) = h(t - k \cdot p_O)\,. \tag{48}$$

The second step consists in the remaining, constellation-centered response, which we write as

$$y_{slr} = \frac{1}{2}\frac{1}{1 - k\cdot n_l} \cdot n_l \cdot \left[h_O^{\rm TT}(t - L - k\cdot p_s^L) - h_O^{\rm TT}(t - k\cdot p_r^L)\right]\cdot n_l\,, \tag{49}$$

where we introduced the positions of the spacecrafts relative to the center of the constellation, $p_A^L \equiv p_A - p_O$.

We will decompose the full signal in the contributions of the individual spin-weighted spherical modes $h_{\ell m}$, whose Fourier transforms are assumed to have a smooth amplitude and phase. First, we define the matrices $P_+, P_\times$ such that, in the sense of matrices,

$$h^{\rm TT} = h_+ P_+ + h_\times P_\times\,. \tag{50}$$

Assuming that the approximation (A6) applies, we focus only on positive frequencies and consider a mode contribution h following (A7). We can define a complex matrix P incorporating the spin-weighted spherical harmonic constant factor as

$$P_{\ell m} = \begin{cases} \frac{1}{2}{}_{-2}Y_{\ell m}\,(P_+ + iP_\times) & \text{for } m > 0\,, \\ \frac{1}{2}{}_{-2}Y^*_{\ell m}\,(P_+ - iP_\times) & \text{for } m < 0\,. \end{cases} \tag{51}$$

We now turn to the transposition of (48) and (49) to the Fourier-domain. For the leading-order response (17), in what we call the local approximation, applying a pure delay as in (48) translates into Fourier-domain transfer function that is a pure phase factor, proportional to the frequency but also t_f-dependent:

$$\mathcal{T}_O^{\rm local}(f) = e^{-2i\pi f d_O(t_f)}\,, \tag{52}$$

with $d_O = -k\cdot p_O$ the orbital delay. When using the leading-order response (17) for the constellation response, the delays take similarly the form of pure phase factors. When also using the rigid approximation for the constellation, where the armlengths are fixed, a particular simplification occurs when combining these individual delays. The Fourier-domain transfer function then reads

$$\mathcal{T}_{slr}^{\rm local} = \frac{i\pi f L}{2}\,\mathrm{sinc}\left[\pi f L\,(1 - k\cdot n_l)\right] \cdot \exp\left[i\pi f\left(L + k\cdot\left(p_1^L + p_2^L\right)\right)\right] n_l \cdot P_{\ell m}\cdot n_l\,, \tag{53}$$

where all vectors n_l and p_A^L are to be evaluated at time t_f. We used the rigid-arms approximation relation $p_r^L - p_s^L = L n_l$. The result (53) is familiar, as it was derived in the past taking the point of view of monochromatic signals.

Note that, if the corrections of Sec. III G are included for the constellation delays, this transfer function will not have this simple form anymore as $\dot{d}$ will have a different velocity-dependent expression for the sending and receiving spacecrafts.

It is natural to define two transfer frequencies for the two relevant length scales for the delays, defined such that a wavelength fits within this length scale. Geometrical projection factors aside, these scales are the radius of the orbit around the Sun $R = 1\text{au} = 1.5\times 10^8$km and the detector armlength $L = 2.5\times 10^6$km [15]. This gives

$$f_R = 3.2\times 10^{-4}\text{Hz} \tag{54a}$$

$$f_L = 1.9\times 10^{-2}\text{Hz}\,. \tag{54b}$$

These transfer frequencies will indicate where in the LISA band a delay term of the form $2\pi f d$ becomes larger than unity. As is well known, for $f > f_L$, the response in (53) starts to acquire a lot of structure and oscillations, as the arguments of the sinc and exp start to be large. Physically, this corresponds to the situation where multiple wavelengths of the signal can be fit within the arm.

B. Estimates for the magnitude of higher-order corrections

Due to the smallness of the annual frequency $\Omega_0/2\pi \simeq 3.1\times 10^{-8}$Hz compared to the timescales relevant for the signal over the LISA frequency band, one can expect that the higher-order corrections described in Section III are small. In this Section we try to quantify to which extent, and give simple estimates of their magnitude.

Using the Newtonian-order expressions (31) for the signal-dependent timescales T_f and T_{A1}, we can obtain a useful order-of-magnitude estimate for the error measures ϵ introduced in (47). Higher-order amplitude terms beyond the first one in (23) will turn out to be completely negligible in the LISA case and we will ignore them in the following. We will present these estimates separately for the orbital response (52) and the constellation (53), as the different baseline of the delays (R or L) as well as the presence of a time-varying prefactor $F(t)$ both affect the result. As we will see, when estimating the magnitude of the relevant derivatives of G we have to take into account delay factors of the type $2\pi f d$.

We start with the orbital response, which takes the form of a pure delay $G_0(f,t) = e^{-2i\pi f d_O(t)}$, and obtain

$$\frac{1}{G_0}\partial_t G_0 = -2i\pi f \dot{d}_O \,, \tag{55a}$$

$$\frac{1}{G_0}\partial_{tt} G_0 = -2i\pi f \ddot{d}_O - 4\pi^2 f^2 \dot{d}_O^2 \,, \tag{55b}$$

$$\frac{1}{G_0}\partial_{tf} G_0 = -2i\pi \dot{d}_O - 4\pi^2 f d_O \dot{d}_O \,, \tag{55c}$$

For the constellation response, since we wish to write errors in relative sense starting from the single-arm response (53) which is already analogous to a discrete time derivative of the signal, we keep explicit the overall factor f reflecting this derivative. We will write symbolically $G_c(f,t) \sim f F(t) e^{-2i\pi f d_c(t)}$, where d_c represents the delay term and $F(t)$ represents the rest of the geometric factors in (53) where we ignore the f-dependence. This gives in this case

$$\frac{1}{G_c}\partial_t G_c \sim -2i\pi f \dot{d}_c + \frac{\dot{F}}{F} \,, \tag{56a}$$

$$\frac{1}{G_c}\partial_{tt} G_c \sim -2i\pi f \ddot{d}_c - 4\pi^2 f^2 \dot{d}_c^2 - 2i\pi f \dot{d}_c \frac{\dot{F}}{F} - \frac{\dot{F}^2}{F^2} + \frac{\ddot{F}}{F} \,, \tag{56b}$$

$$\frac{1}{G_c}\partial_{tf} G_c \sim -4i\pi \dot{d}_c - 2i\pi d_c \frac{\dot{F}}{F} - 4\pi^2 f d_c \dot{d}_c + \frac{1}{f}\frac{\dot{F}}{F} \,. \tag{56c}$$

In order to obtain simple scalings for the error estimates, we will make the replacements $\partial_t^n d \sim \Omega_0^n d$ as well as $\partial_t^n F \sim \Omega_0^n F$. These scalings are only approximate and the various possible orientation angles can lead to significant variations. We will assume $d_0 \sim R_{\rm eff}$ (recalling that we set $c = 1$) with an effective orbit radius $R_{\rm eff} \equiv 2R/\pi$ representing an average of the geometric projection factor of the gravitational wave propagation vector on the plane of the orbit. For the constellation delays, we simply take $d_c \sim L$ as there projection effects can lead to variations in both ways. We build estimates for the magnitude of these derivatives as follows:

$$\left|\frac{1}{G_0}\partial_t G_0\right| \sim 2\pi f \Omega_0 R_{\rm eff} \,, \tag{57a}$$

$$\left|\frac{1}{G_0}\partial_{tt} G_0\right| \sim \max\left[2\pi f \Omega_0^2 R_{\rm eff}, 4\pi^2 f^2 \Omega_0^2 R_{\rm eff}^2\right] \,, \tag{57b}$$

$$\left|\frac{1}{G_0}\partial_{tf} G_0\right| \sim \max\left[2\pi \Omega_0 R_{\rm eff}, 4\pi^2 f \Omega_0 R_{\rm eff}^2\right] \,, \tag{57c}$$

for the orbital response and

$$\left|\frac{1}{G_c}\partial_t G_c\right| \sim \max\left[\Omega_0, 2\pi f \Omega_0 L\right] \,, \tag{58a}$$

$$\left|\frac{1}{G_c}\partial_{tt} G_c\right| \sim \max\left[\Omega_0^2, 4\pi f \Omega_0^2 L, 4\pi^2 f^2 \Omega_0^2 L^2\right] \,, \tag{58b}$$

$$\left|\frac{1}{G_c}\partial_{tf} G_c\right| \sim \max\left[\Omega_0/f, 6\pi \Omega_0 L, 4\pi^2 f \Omega_0^2 L^2\right] \,, \tag{58c}$$

for the constellation response. In the presence of different terms, we simply take the maximum of their norms. An important point in the above is the presence of delay factors in the forms of powers of $2\pi f d$. Thus, the magnitude of the corrections cannot be evaluated with the simple replacement $\partial_t^n \to \Omega_0^n$, and the frequency dependency of the error measures will depend on the frequency being above or below the transfer frequencies f_R and f_L defined in (54).

The error measure ϵ_d does not depend on signal-dependent timescales and can be directly read off the estimates above, with $\epsilon_d = \partial_{tf} G/(2\pi G)$. For the error measures ϵ_Ψ and ϵ_{A1}, we can combine the above with the Newtonian timescales given in (31) for the inspiral phase of the signal. An important distinction needs to be made there, as the radiation reaction timescale T_f grows rapidly towards lower frequencies as $f^{-11/6}$. If one considers systems that are arbitrarily far from merging, T_f can be made arbitrarily large, leading the error measure ϵ_Ψ to take values arbitrarily larger than 1 where the perturbative treatment we adopted breaks down. This limit would describe systems observed in their deep inspiral and very slowly chirping, such as the population of galactic binaries that form one of the science targets of the LISA mission [15, 19]. Such signals would be very compact in Fourier domain, and are not covered by our formalism that assumes an extended Fourier spectrum. On the contrary, if one considers only systems that will merge during or shortly after the observation span of the mission, the starting frequency of the observations will be set by this limited lifetime of the source before merging and we can then bound ϵ_Ψ to be smaller than 1.

In the following we will consider only these merging binaries, which include the supermassive black hole mergers that we expect to detect. The population of stellar-origin binary black holes, proposed in [19], will have both sources rather far from merger and sources that will merge within a few years after the mission finishes. We will take a reference starting frequency corresponding to a time from merger $\Delta t = 10$yrs. Since the corrections

turn out to be most important at the beginning of the signal, we will evaluate the timescales at this starting frequency. From the Newtonian relation (7), we have (using $\nu_0 = 1/4$ for the equal-mass value):

$$f_{\rm start} = 2.9\times10^{-5}{\rm Hz}\left(\frac{\nu}{\nu_0}\right)^{-3/8}\left(\frac{m}{10^6 M_\odot}\right)^{-5/8}\left(\frac{\Delta t}{10{\rm yr}}\right)^{-3/8} \tag{59}$$

which gives in turn for the timescales

$$T_f = 6.8\times10^{-2}{\rm yr}\left(\frac{\nu}{\nu_0}\right)^{3/16}\left(\frac{m}{10^6 M_\odot}\right)^{5/16}\left(\frac{\Delta t}{10{\rm yr}}\right)^{11/16} \tag{60a}$$

$$T_{A1} = 2.0\times10^{-4}{\rm yr}\left(\frac{\nu}{\nu_0}\right)^{3/8}\left(\frac{m}{10^6 M_\odot}\right)^{5/8}\left(\frac{\Delta t}{10{\rm yr}}\right)^{3/8}. \tag{60b}$$

When inserting these expressions in (47), in (57) and (58) we must also use $f_{\rm start}$ for the possible extra factors of f or f^2 coming from the delay terms, which can lead to different overall scalings with mass ratio and total mass depending on where the signal starts in the band.

We summarize the results for the error estimates in Fig. 3. We show the magnitude of the figures of merit ϵ_Ψ, ϵ_{A1}, ϵ_d both for the orbital delay and the constellation response, for an equal-mass system with total masses of $m = 10^7 M_\odot$, $10^4 M_\odot$ and $10^2 M_\odot$ and a starting frequency corresponding to $\Delta t = 10$yr of observation. Since individual signals can show significant variation depending on the angular parameters, we show an average of $\ln\epsilon$ over the sky position, inclination and polarization, together with $\pm\sigma$ standard deviation, and overlay the estimates resulting from (57), (58) and (31). Here and in the following, for completeness we show results on the frequency band from 10^{-5}Hz to $1Hz$, while the signals are expected to have little signal-to-noise ratio outside of the band from 10^{-4}Hz to $10^{-1}Hz$, which is sometimes taken as a reference [15].

For ϵ_Ψ, we find as expected a steep rise towards lower frequencies, which shows that considering only merging binaries with a limited time remaining before merger is crucial here. Due to the different occurences of delay factors, we find a different behaviour between the orbital response, where the initial value of ϵ_Ψ grows towards lower masses, and the constellation response, where ϵ_Ψ grows towards higher masses. In both cases, the error measure remains below $\epsilon_\Psi \sim 0.1$, where the corrections can be sizeable but well manageable by the perturbative formalism (as we will show in the next Section). Not represented in the figure is the behaviour with mass ratio, which also differs: for a total mass of $10^2 M_\odot$, the extra factors of $f_{\rm start}$ due to the delay factors lead to an increase towards larger mass ratios, while for a total mass of $10^7 M_\odot$ we have a decrease towards larger mass ratios.

For ϵ_{A1}, we find that the structure in the waveform close to merger does have a visible but limited impact on the error measure. For the orbital response, the error measure remains small for all masses. For the constellation response, we find that ϵ_{A1} can grow up to ~ 0.01 at the lower end of the frequency band for high masses. Since the timescale $T^{\rm N}_{A1}$ has a simple dependence $\propto 1/f$, these statements should hold for all mass ratios at least during the inspiral.

, The ϵ_d is signal-independent, and follows a quite different behaviour for the orbital and constellation responses. For the orbital response, ϵ_{d_0} grows for higher frequencies, reaching the ~ 0.1 at around 1Hz. For the constellation response, ϵ_{d_c} is mainly important at lower frequencies, reaching ~ 0.01 at around 10^{-5}Hz.

Overall, we find that for merging binaries the perturbative approach presented in Sec. III will apply, with the most relevant higher-order corrections being expected to be the phase corrections close to the starting frequency of the signal and the delay correction at high frequencies for the orbital response.

C. Error control for the Fourier-domain response

Having at hand generic estimates for where the higher-order corrections in the LISA response are expected to be relevant, we now demonstrate that including them indeed reduces the reconstruction error of the Fourier-domain waveform. We will use an example PhenomD [3] waveform for an equal-mass and non-spinning system, and consider only two values for the total mass, $m = 10^7 M_\odot$ and $m = 10^2 M_\odot$. For this example, we will also focus on the orbital response and on a single-arm constellation response y_{132}, as TDI combinations will appear (within the approximations listed above) as linear combinations of these basic observables. On one hand, we perform a numerical inverse Fourier transform (IFFT) of the Fourier-domain PhenomD waveform, process the signal through the time-domain response (48) or (49), and perform a numerical Fourier transform (FFT). This gives us the target Frourier-domain waveform, considered to be exact but for the possible artifacts of numerical Fourier transforms (most notably, oscillations induced by the necessary tapering of the signal). On the other hand, we process the Fourier-domain signal through the response summarized in Sec. III H, including various higher-order corrections. We then compare the output of the two procedures.

We show the results for the orbital response in Figs. 4 and 5 in amplitude and phase form. In both cases, in accordance with (52), the transfer function is essentially a phase, and contains many more cycles in the low-mass case in the higher frequency band. We find that including higher-order corrections does reduce the reconstruction errors, but with the distincitve feature that including only one of the amplitude and delay corrections can make the approximation actually worse, while including both does improve the accuracy. In the high-mass case, we also find for all orders of approximation an increase in the errors at the very end. This is likely to be due to the fact that the errors in the transfer function are to be in-

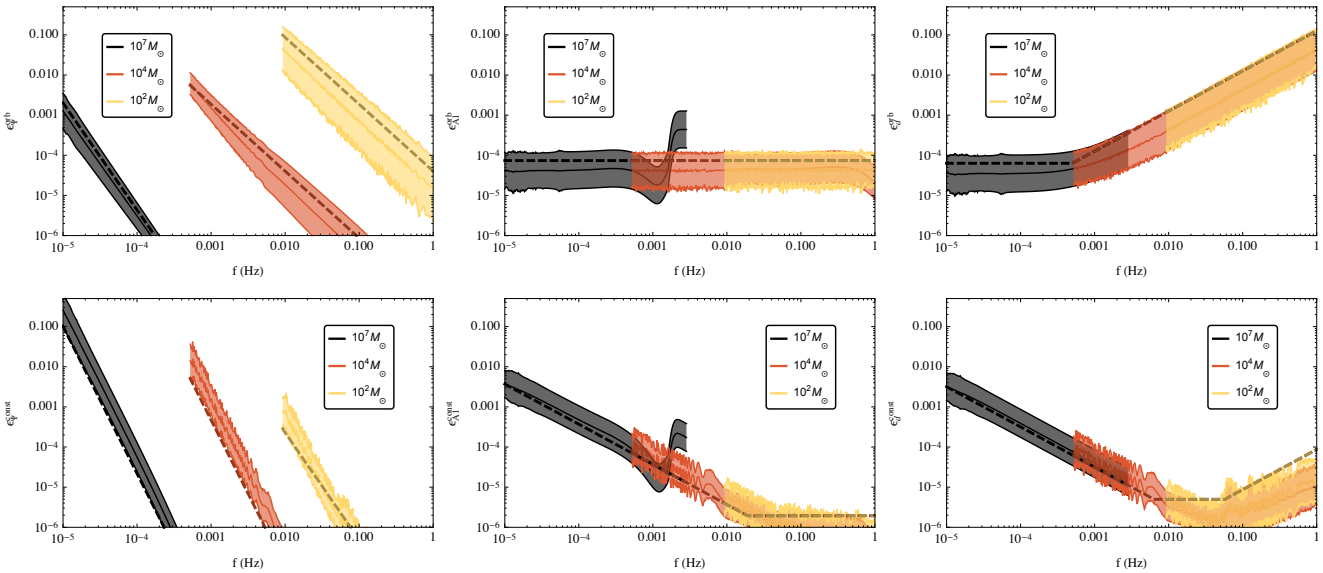


FIG. 3. Error estimates of the approximation as defined in (47), for an equal-mass, non-spinning system and for total masses of $10^7 M_\odot$, $10^4 M_\odot$ and $10^2 M_\odot$. The top row corresponds to the orbital delay part of the response (48), and the lower row shows the LISA-centered constellation response (49). The error estimates (47) are shown for the phase, amplitude and delay terms from left to right. The central line and interval are the mean and 1σ standard deviation of the logarithm of ϵ over 400 random values for the position in the sky, inclination and polarization. The starting frequency is set by an observation time of $\Delta t = 10$yrs before merger.

terpreted in the relative sense, and the amplitude of the signal is decaying rapidly in this post-merger region. Although we did not identify the precise cause of this rise in errors, one possible explanation would be tapering-induced oscillations in the numerical Fourier transforms.

For the constellation response, we show the results in Figs. 6 and 7, using directly the real and imaginary part of the transfer function. We find similar features as for the orbital response: including higher-corrections reduces the errors, but in the high-mass case the amplitude and delay corrections should be added together. We also find the same uncontrolled grow of errors at the very end of the signal in the high-mass case, in the region where the amplitude of the signal decays.

Acknowledgments

[Acknowledgments]

Appendix A: Notation and conventions

The convention we will be using for the Fourier transform of a signal $h(t)$ is

$$\tilde{h}(f) = \mathrm{FT}[h](f) = \int \mathrm{d}t\, e^{+2i\pi f t} h(t)\,. \tag{A1}$$

Notice that this sign convention is not the most frequently used in the literature. We chose it to ensure that, with the conventions of [17], spin-weighted spherical modes $h_{\ell m}$ with $m > 0$ will have support mostly for positive frequencies. One can revert to the more usual convention by taking $f \to -f$.

A useful representation of the gravitational waveform is given by its decomposition in spin-weighted spherical harmonics. The waveform emitted in the direction (Θ, Φ), with its two polarizations $h_+, h_\times$, can be decomposed as a superposition of modes as [20]

$$h_+ - i h_\times = \sum_{\ell \geq 2} \sum_{m=-\ell}^{\ell} {}_{-2}Y_{\ell m}(\Theta, \Phi) h_{\ell m}\,, \tag{A2}$$

where the ${}_{-2}Y_{\ell m}(\Theta, \Phi)$ are spin-weighted spherical harmonics [21]. Conversely, the individual polarizations are obtained from the individual modes as

$$h_+ = \frac{1}{2} \sum_{\ell, m} \left[{}_{-2}Y_{\ell m} h_{\ell m} + {}_{-2}Y^*_{\ell m} h^*_{\ell m} \right]\,, \tag{A3a}$$

$$h_\times = \frac{i}{2} \sum_{\ell, m} \left[{}_{-2}Y_{\ell m} h_{\ell m} - {}_{-2}Y^*_{\ell m} h^*_{\ell m} \right]\,. \tag{A3b}$$

For a non-precessing system, with a fixed orbital plane, the individual modes have the additional symmetry property

$$h_{\ell, -m} = (-1)^\ell h^*_{\ell m}\,. \tag{A4}$$

Since we will work in the Fourier domain, it will be useful to write the contributions of the individual modes to the

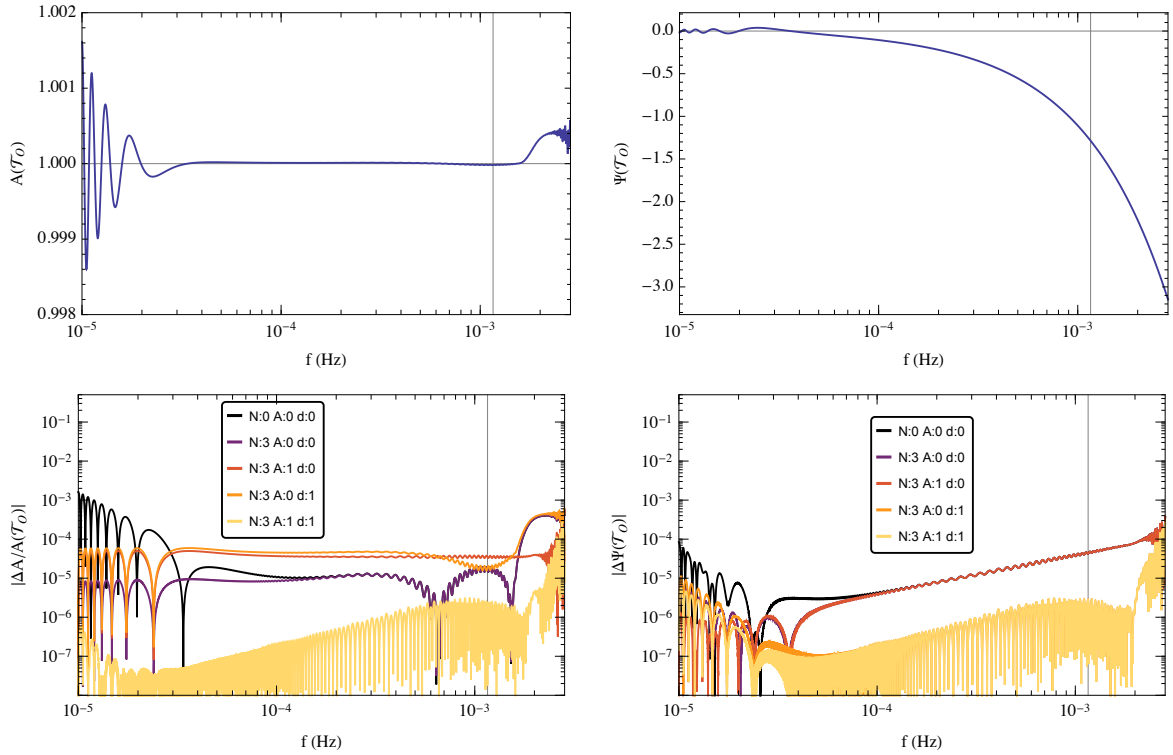


FIG. 4. Error in the transfer function for the orbital response, for $M = 10^7 M_\odot$ (equal mass, non-spinning). The top panels show the amplitude and phase of the transfer function. The residuals are shown in the bottom panels separately for the amplotude and phase, for various orders of approximation as outlined in Sec. III H

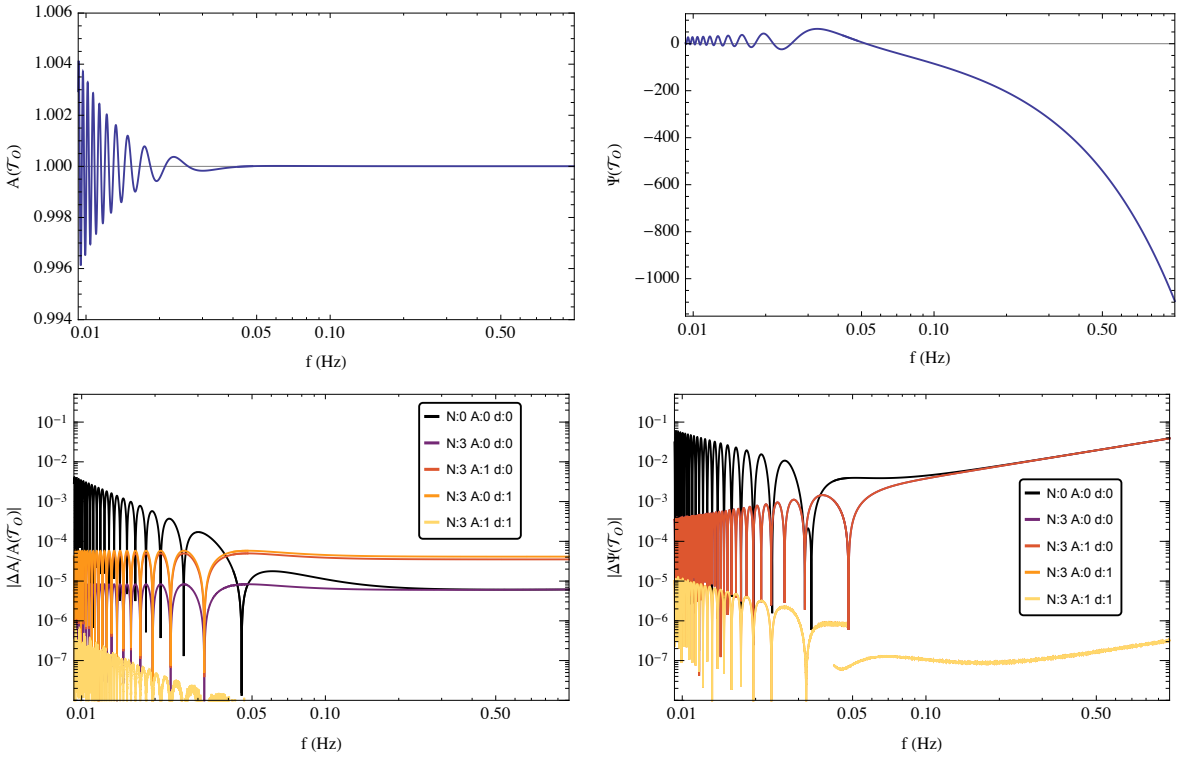


FIG. 5. Transfer function and reconstruction error for the orbital response, for $M = 10^2 M_\odot$.

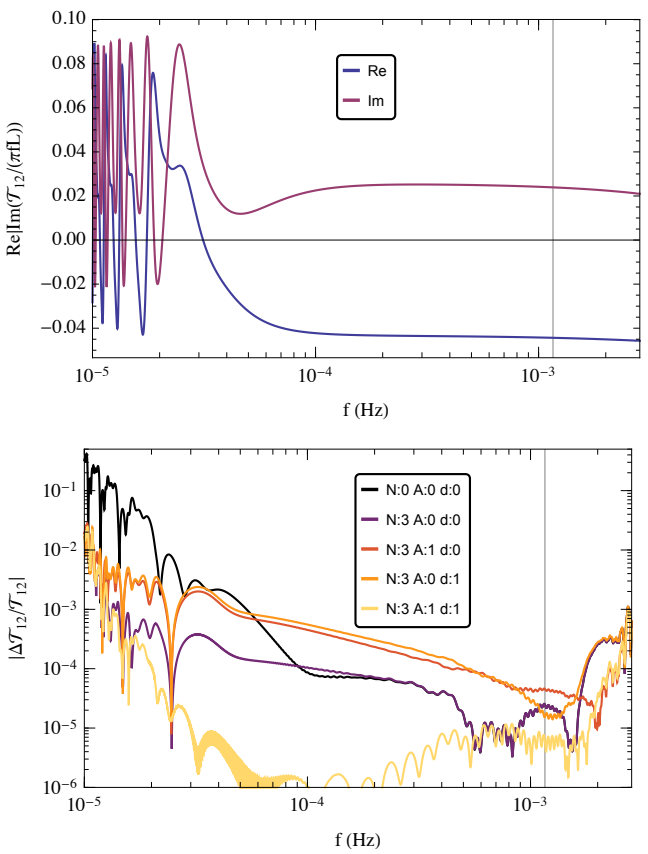


FIG. 6. Transfer function and reconstruction error for the basic observable y_{132}, for $M = 10^7 M_\odot$. Here the top panels show directly the real and imaginary part of the transfer function, and the bottom panel shows the norm of the residuals.

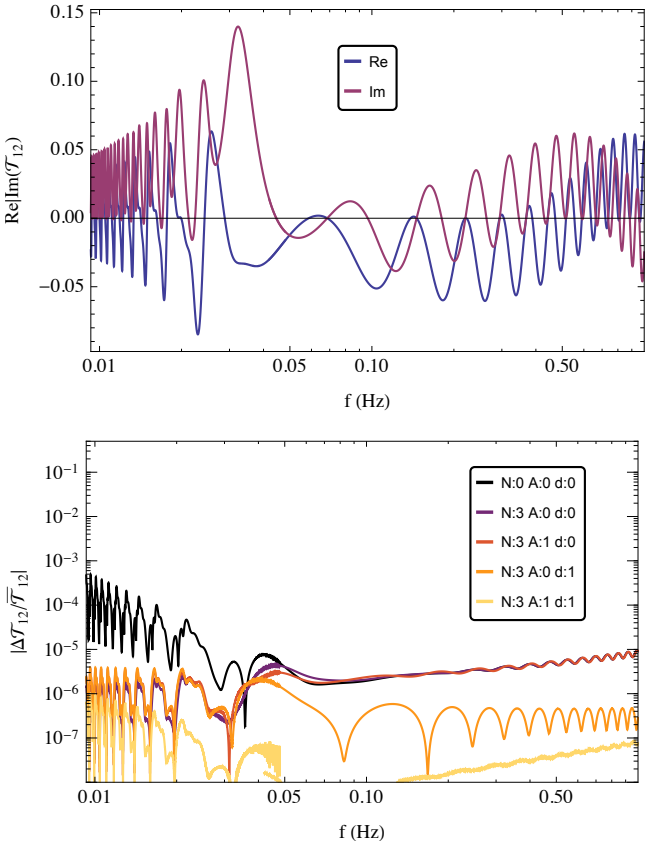


FIG. 7. Transfer function and reconstruction error for the basic observable y_{132}, for $M = 10^2 M_\odot$. Here the top panels show directly the real and imaginary part of the transfer function, and the bottom panel shows the norm of the residuals.

Fourier transforms of the polarizations as

$$\tilde{h}_+(f) = \frac{1}{2}\sum_{\ell\geq 2}\sum_{m=-\ell}^{\ell}\left[{}_{-2}Y_{\ell m}\tilde{h}_{\ell m}(f) + {}_{-2}Y^*_{\ell m}\tilde{h}_{\ell m}(-f)^*\right], \tag{A5a}$$

$$\tilde{h}_\times(f) = \frac{i}{2}\sum_{\ell\geq 2}\sum_{m=-\ell}^{\ell}\left[{}_{-2}Y_{\ell m}\tilde{h}_{\ell m}(f) - {}_{-2}Y^*_{\ell m}\tilde{h}_{\ell m}(-f)^*\right], \tag{A5b}$$

where we used $\widetilde{h^*_{\ell m}}(f) = \tilde{h}_{\ell m}(-f)^*$. An additional approximation often used in waveform models consists in considering that the Fourier transforms $\tilde{h}_{\ell m}$ have support only on one side of the spectrum, either for positive of for negative frequencies depending on the sign of m. This holds in particular within the stationary phase approximation, as explained below. With our sign convention (A1), this approximation reads

$$\begin{aligned}\tilde{h}_{\ell m}(f) &\simeq 0 \text{ for } f<0,\ m>0\,,\\ \tilde{h}_{\ell m}(f) &\simeq 0 \text{ for } f>0,\ m<0\,.\end{aligned} \tag{A6}$$

In the following, we will drop the mode indices and the sum over modes, and we will denote generically by h an individual mode of the waveform[3]. Note that, if we are considering real signals $s(t)$ (such as h_+, $h_\times$, or any detector output), we can focus on the positive-frequencies range and reconstruct the negative frequencies as $\tilde{s}(-f) = \tilde{s}(f)^*$. For positive frequencies $f > 0$, we will use the generic notation h as

$$\begin{aligned}h(t) &\equiv h_{\ell m}(t)\,, \quad \tilde{h}(f) \equiv \tilde{h}_{\ell m}(f) \text{ for } m>0\,,\\ h(t) &\equiv h^*_{\ell m}(t)\,, \quad \tilde{h}(f) \equiv \tilde{h}_{\ell m}(-f)^* \text{ for } m<0\,.\end{aligned} \tag{A7}$$

We will further decompose $\tilde{h}$ into a Fourier-domain amplitude A and a phase Ψ according to

$$\tilde{h}(f) \equiv A(f)e^{-i\Psi(f)}\,. \tag{A8}$$

Since we intend to work in the Fourier domain, we will use PhenomD waveforms [3]. These phenomenological waveforms provide analytic fits for both the Fourier-domain amplitude and phase of the 22 mode, with fit

[3] This notation for a generic individual mode is not to be confused with the often encountered notation $h \equiv h_+ - ih_\times$.

coefficients that were calibrated on EOB/NR hybrids [describe smoothing across intervals].

We recall that, with our sign convention, the effect of a positive shift in time $h'(t) = h(t - \Delta t)$ is

$$\tilde{h}'(f) = e^{2i\pi f \Delta t}\tilde{h}(f)\,. \tag{A9}$$

If we also allow for a global shift in phase of the signal, which corresponds to a rotation of the unit vector pointing towards the observer in the frame of the gravitational waves source, a constant term can also be added (note that the effect of such a rotation gives different phase changes $e^{\pm im\delta\phi}$ for the different modes of the radiation). We see that both a constant and a linear term in the Fourier-domain phase can be adjusted by changing extrinsic parameters. Thus, as is well known, apart from the relative phases of the modes for waveforms that include higher harmonics, the intrinsic information on the phasing is contained in the second derivative of the Fourier-domain phase $\Psi(f)$.

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